

CHEMISTRY CHAPTER 17

CHEMICAL EQUILIBRIUM

COMPLETE ANSWER KEYS

Includes verified answer keys for:

- Quiz 1 — Equilibrium Fundamentals (conceptual)
- Quiz 2 — Le Châtelier's Principle (conceptual)
 - Quiz 3 — Keq Calculations (math)
- Quiz 4 — Ksp & Solution Equilibrium (math + concept)
 - Quiz 5 — Comprehensive Review (mixed)

*All answers verified against Chapter 17 PPT source material.
All math problems show complete step-by-step worked solutions.*

ANSWER KEY — Quiz 1: Equilibrium Fundamentals

Multiple Choice

1. b. forward and reverse rates are equal AND concentrations are stable (not necessarily equal)
2. b. Haber process $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ is the textbook example of a reversible reaction
3. b. $K_{eq} = [\text{C}]^c[\text{D}]^d / [\text{A}]^a[\text{B}]^b$ — products over reactants, coefficients as exponents
4. c. pure solids and liquids are defined as concentration = 1, so they are excluded
5. b. $K_{eq} = 650 > 1 \rightarrow$ forward reaction predominates, mostly products present

True / False

6. True — both reactants and products coexist in a reversible reaction mixture
7. False — concentrations are stable (constant) at equilibrium, but NOT necessarily equal
8. True — equilibrium position depends on conditions such as temperature and concentration
9. False — $K_{eq} < 1$ means the REVERSE reaction predominates; $[\text{reactants}] > [\text{products}]$
10. True — K_{eq} must always be determined experimentally — stated directly in the PPT
11. False — an irreversible reaction goes in the FORWARD direction ONLY
12. True — homogeneous = all substances in the same state of matter
13. False — pure liquids are NOT included — their concentration is defined as 1 and drops out
14. True — definition of chemical equilibrium: forward rate = reverse rate
15. True — $K_{eq} > 1 \rightarrow$ equilibrium lies right $\rightarrow [\text{products}] > [\text{reactants}]$

Short Answer

16.

Reaction: $3\text{H}_2(\text{g}) + \text{N}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$

$K_{eq} = [\text{NH}_3]^2 / ([\text{N}_2][\text{H}_2]^3)$

Products in numerator, reactants in denominator, coefficients as exponents.

All substances are gases $\rightarrow$ homogeneous equilibrium $\rightarrow$ all included.

17.

Reversible reaction: can occur in BOTH forward and reverse directions.

Example: $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ (Haber process) — products can reform reactants.

Irreversible reaction: occurs in the FORWARD direction only.

Example: burning wood — products cannot convert back to wood.

18.

$K_{eq} = 0.0025 < 1 \rightarrow$ reverse reaction predominates.

Equilibrium lies to the LEFT.

The reaction is REACTANT-FAVORED — very little product forms at equilibrium.

Reactant concentration is much greater than product concentration.

Essay — Key Points

19. — Key points for full credit:

- Chemical equilibrium = forward and reverse reaction rates are equal.
- Concentrations stop CHANGING (stable), but forward/reverse reactions do NOT stop.
- Both reactions continue occurring simultaneously at the same rate.
- Equilibrium is dynamic, not static — molecules are still reacting.

20. — Key points for full credit:

- Law of chemical equilibrium: at a specific T, ratio of [products]/[reactants] is constant.
- Write Keq: products in numerator, reactants in denominator.
- Use coefficients from balanced equation as exponents.
- Exclude pure solids and pure liquids (concentration defined as 1).
- Example: $3\text{H}_2 + \text{N}_2 \rightleftharpoons 2\text{NH}_3 \rightarrow \text{Keq} = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$

21. — Key points for full credit:

- Homogeneous: all substances same phase. Example: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, all gases.
- $\text{Keq} = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$ — all included.
- Heterogeneous: substances in different phases. Example: $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$.
- $\text{Keq} = [\text{CO}_2]$ — solids excluded.
- Solids/liquids excluded because their 'concentration' is constant = 1 (doesn't change).

ANSWER KEY — Quiz 2: Le Châtelier's Principle

Multiple Choice

1. b. Le Châtelier: system proceeds in direction that RELIEVES the stress
2. b. increasing [reactant] → shift right (forward) to consume the added reactant
3. b. increasing pressure → shift toward side with FEWER moles of gas
4. c. temperature is the ONLY stress that changes the value of K_{eq}
5. b. exothermic NH_3 : increasing T → like adding product → shift LEFT → K_{eq} decreases

True / False

6. True — concentration changes shift position but do NOT change K_{eq} — confirmed by PPT table
7. False — pressure shifts toward FEWER moles of gas — not always to the right
8. False — catalyst speeds approach to equilibrium but does NOT shift position or change K_{eq}
9. True — removing product → shift right to produce more product (Le Châtelier)
10. True — for exothermic reaction, thermal energy is a product, so adding T = adding product
11. False — decreasing pressure → shift toward MORE moles of gas, not fewer
12. False — solids do not appear in K_{eq} expression → adding solid has NO effect on equilibrium
13. True — catalysts affect RATE only, not equilibrium position or K_{eq} value
14. True — endothermic: thermal energy is a reactant; increasing T = adding reactant → shift right → K_{eq} increases
15. True — removing reactant → shift LEFT to replace the lost reactant

Short Answer

16.

Reaction: $CO(g) + 3H_2(g) \rightleftharpoons CH_4(g) + H_2O(g)$ (exothermic)

Left side: $1+3 = 4$ mol gas. Right side: $1+1 = 2$ mol gas.

- a. Pressure increased → shift RIGHT (toward fewer moles: 2 vs 4)
- b. Temperature increased → like adding product (exothermic) → shift LEFT, K_{eq} decreases
- c. CO removed → shift LEFT to replace lost reactant
- d. H_2O added → shift LEFT to consume added product

17.

Le Châtelier's principle: when a stress is placed on a reversible process, it proceeds in the direction that relieves that stress.

Industries (e.g., Haber process for NH_3) manipulate pressure, temperature, and concentration to drive the reaction toward the desired product, maximizing yield.

Example: high pressure + low temperature + removing NH_3 as it forms → maximizes NH_3 production.

18.

The student is **INCORRECT**.

A catalyst speeds up **BOTH** the forward and reverse reactions equally.

This means equilibrium is reached more quickly, but the equilibrium **POSITION** does not shift.

The value of K_{eq} does not change. The concentrations at equilibrium are unchanged.

Only the **TIME** to reach equilibrium is reduced, not the final equilibrium state.

Essay — Key Points

19. — Key points for full credit:

- Concentration: increasing [reactant] → shift right; removing product → shift right. K_{eq} unchanged.
- Pressure (gases only): increase P → shift toward fewer moles of gas. K_{eq} unchanged.
- Temperature: exothermic + increase T → shift left, K_{eq} decreases. Endothermic + increase T → shift right, K_{eq} increases.
- Only temperature changes K_{eq} because thermal energy **IS** a reactant or product; concentration/pressure are not.

20. — Key points for full credit:

- Temperature changes the equilibrium position because heat is treated as a reactant (endothermic) or product (exothermic).
- Adding thermal energy → like changing concentration of heat → shifts equilibrium position.
- Since thermal energy is **NOT** in the K_{eq} equation itself, the new equilibrium position produces a different ratio of [products]/[reactants] → different K_{eq} .
- Concentration/pressure changes only shift position; the same K_{eq} is eventually re-established.

21. — Key points for full credit:

- Reaction: $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (exothermic, 4 mol gas → 2 mol gas).
- Temperature: use **LOW** temperature → favors forward (exothermic), increases K_{eq} . Trade-off: slower rate.
- Pressure: use **HIGH** pressure → shifts toward fewer moles (products side). No K_{eq} change.
- Concentration: continuously **REMOVE** NH_3 as it forms → equilibrium keeps shifting right.
- Add excess N_2 or H_2 → pushes equilibrium right.
- Real industry uses moderate T with catalyst for acceptable rate + yield balance.

ANSWER KEY — Quiz 3: Keq Calculations

Multiple Choice

1. a. $K_{eq} = [\text{NO}_2]^2 / ([\text{NO}]^2[\text{O}_2])$ — products over reactants, coefficients as exponents
2. b. CaCO_3 and CaO are solids (excluded) $\rightarrow K_{eq} = [\text{CO}_2]$ only
3. c. Mg is solid (excluded); $K_{eq} = [\text{MgCl}_2][\text{H}_2]/[\text{HCl}]^2 = (0.513 \times 0.411)/(0.051)^2 \approx 81$
4. a. $K_{eq} = (0.0329)^2 / ((0.897)^2 \times 0.254) = 1.082 \times 10^{-3} / 2.041 \times 10^{-1} \approx 5.30 \times 10^{-3}$
5. b. error: coefficients 2 (on NH_3) and 3 (on H_2) must be used as exponents

True / False

6. True — exponents in K_{eq} = coefficients in balanced equation — fundamental rule
7. False — pure solids are EXCLUDED entirely from K_{eq} (not given concentration = 1 in the expression)
8. False — large K_{eq} means NUMERATOR (products) is larger than denominator (reactants)
9. True — K_{eq} numerical value must be measured experimentally — stated in PPT
10. True — $K_{eq} = [\text{C}][\text{D}]/[\text{A}][\text{B}] = (1 \times 1)/(2 \times 2) = 0.25$ ✓
11. False — K_{eq} is a ratio; doubling all concentrations keeps the ratio the same — K_{eq} unchanged
12. True — aqueous ions ARE included in K_{eq} expressions (only solids and pure liquids excluded)
13. True — homogeneous = all species same phase $\rightarrow$ all included, same phase throughout
14. True — K_{eq} is dimensionless — units cancel in the ratio
15. True — [products] > [reactants] at equilibrium $\rightarrow$ numerator > denominator $\rightarrow K_{eq} > 1$

Calculations

16. (Example 17-3)

Reaction: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$

Known: $[\text{N}_2]=0.045 \text{ M}$, $[\text{H}_2]=0.055 \text{ M}$, $[\text{NH}_3]=52 \text{ M}$

Formula: $K_{eq} = [\text{NH}_3]^2 / ([\text{N}_2][\text{H}_2]^3)$

$$(52)^2 / (0.045 \times (0.055)^3)$$

$$2704 / (0.045 \times 1.664 \times 10^{-4})$$

$$2704 / 7.49 \times 10^{-6}$$

$$K_{eq} \approx 3.61 \times 10^8$$

17.

Reaction: $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$

Known: $[\text{NO}]=0.897 \text{ M}$, $[\text{O}_2]=0.254 \text{ M}$, $[\text{NO}_2]=0.0329 \text{ M}$

Formula: $K_{eq} = [\text{NO}_2]^2 / ([\text{NO}]^2[\text{O}_2])$

$$(0.0329)^2 / ((0.897)^2 \times 0.254)$$

$$1.082 \times 10^{-3} / (0.8046 \times 0.254)$$

$$1.082 \times 10^{-3} / 0.2044$$

$$K_{eq} \approx 5.296 \times 10^{-3}$$

18.



Mg is a pure solid → EXCLUDED from K_{eq} expression.

K_{eq} expression: $K_{eq} = [MgCl_2][H_2] / [HCl]^2$

Known: $[HCl]=0.051\text{ M}$, $[MgCl_2]=0.513\text{ M}$, $[H_2]=0.411\text{ M}$

$(0.513 \times 0.411) / (0.051)^2$

$0.2108 / 0.002601$

$K_{eq} \approx 81.1$

Essay — Key Points

19. — Key points for full credit:

- Step 1: Write what you know (list all equilibrium concentrations).
- Step 2: Write the K_{eq} expression from the balanced equation (products/reactants, coefficients as exponents, exclude solids/liquids).
- Step 3: Plug in known values.
- Step 4: Evaluate (calculate).
- Example (17-3): $K_{eq} = [NH_3]^2 / ([N_2][H_2]^3) = (52)^2 / (0.045 \times (0.055)^3) \approx 3.61 \times 10^8$

20. — Key points for full credit:

- Rule 1: Products in numerator, reactants in denominator.
- Rule 2: Coefficients from balanced equation become exponents.
- Rule 3: Pure solids and pure liquids are excluded (concentration = 1, constant).
- Exponents represent how many moles of each species participate.
- K_{eq} must be determined experimentally because it depends on actual measured concentrations at equilibrium for a specific reaction at a specific temperature.

21. — Key points for full credit:

- $K_{eq} = 8.2 \times 10^{-10}$: extremely small → heavily reactant-favored → barely any product forms.
- $K_{eq} = 4.5 \times 10^5$: extremely large → heavily product-favored → reaction goes nearly to completion.
- The 4.5×10^5 reaction is far more product-favored.
- 8.2×10^{-10} : equilibrium lies far LEFT; $[reactants] \gg [products]$.
- 4.5×10^5 : equilibrium lies far RIGHT; $[products] \gg [reactants]$.

ANSWER KEY — Quiz 4: K_{sp} & Solution Equilibrium

Multiple Choice

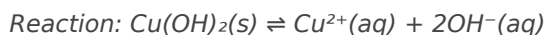
1. a. K_{sp} = equilibrium constant for dissolving a solute into a saturated solution (PPT def.)
2. c. $\text{AgCl(s)} \rightleftharpoons \text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow K_{\text{sp}} = [\text{Ag}^+][\text{Cl}^-]$ (solid excluded)
3. c. ion product > K_{sp} → supersaturated (too many ions → precipitate forms)
4. b. common-ion effect: common ion from two solutions must be considered in solubility calculations
5. c. $[\text{Cu}^{2+}] = S$, $[\text{OH}^-] = 2S \rightarrow K_{\text{sp}} = S \times (2S)^2 = 4S^3$

True / False

6. True — K_{sp} > 1 → large number of ions in solution → soluble (PPT)
7. False — very small K_{sp} → very FEW ions in solution → insoluble, not highly soluble
8. True — common ion shifts equilibrium left (Le Châtelier) → less dissolves → precipitate forms
9. True — ion product = K_{sp} → solution is exactly saturated (equilibrium)
10. True — compare ion concentration product to K_{sp} to predict precipitation
11. False — the solid is excluded from K_{sp} expression (concentration defined as 1)
12. True — in pure water, $\text{AgCl} \rightarrow \text{Ag}^+ + \text{Cl}^-$: one Ag⁺ and one Cl⁻ per formula unit → $[\text{Ag}^+] = [\text{Cl}^-]$
13. True — ion product > K_{sp} → supersaturated → precipitate forms until K_{sp} is reached
14. True — K_{sp} can be used to calculate solubility in mol/L (Example 17-7)
15. False — common-ion effect DECREASES solubility (equilibrium shifts left, less dissolves)

Calculations

16. (Example 17-6)



$$\text{Solubility } S = 1.76 \times 10^{-7} \text{ mol/L}$$

$$[\text{Cu}^{2+}] = S = 1.76 \times 10^{-7} \text{ M}$$

$$[\text{OH}^-] = 2S = 3.52 \times 10^{-7} \text{ M}$$

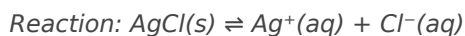
$$K_{\text{sp}} = [\text{Cu}^{2+}][\text{OH}^-]^2 = S \times (2S)^2 = 4S^3$$

$$4 \times (1.76 \times 10^{-7})^3$$

$$4 \times 5.452 \times 10^{-21}$$

$$K_{\text{sp}} \approx 2.18 \times 10^{-20}$$

17. (Example 17-7)



$$K_{\text{sp}} = 1.8 \times 10^{-10}$$

$$[\text{Ag}^+] = [\text{Cl}^-] = S$$

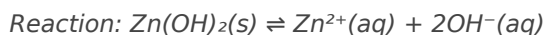
$$K_{\text{sp}} = [\text{Ag}^+][\text{Cl}^-] = S \times S = S^2$$

$$S^2 = 1.8 \times 10^{-10}$$

$$S = \sqrt{(1.8 \times 10^{-10})}$$

$$\text{Solubility} \approx 1.34 \times 10^{-5} \text{ mol/L}$$

18. (Example 17-8)



$$[\text{Zn(OH)}_2] = 2.5 \times 10^{-6} \text{ M}$$

$$[\text{Zn}^{2+}] = [\text{Zn(OH)}_2] = 2.5 \times 10^{-6} \text{ M}$$

$$[\text{OH}^-] = 2 \times [\text{Zn(OH)}_2] = 5.0 \times 10^{-6} \text{ M}$$

$$\text{Ion concentration product} = [\text{Zn}^{2+}][\text{OH}^-]^2$$

$$(2.5 \times 10^{-6})(5.0 \times 10^{-6})^2$$

$$(2.5 \times 10^{-6})(2.5 \times 10^{-11})$$

$$6.25 \times 10^{-17}$$

$$K_{sp} = 3.0 \times 10^{-17}$$

Ion product $6.25 \times 10^{-17} > K_{sp} 3.0 \times 10^{-17} \rightarrow \text{SUPERSATURATED}$

Essay — Key Points

19. — Key points for full credit:

- K_{sp} = solubility product constant = K_{eq} for dissolving a solute in a saturated solution.
- $K_{sp} > 1$: large number of ions $\rightarrow$ soluble. $K_{sp} < 1$: few ions $\rightarrow$ insoluble.
- To predict precipitation: calculate ion concentration product for the mixed solution.
- If ion product $> K_{sp} \rightarrow$ precipitate forms. If $< K_{sp} \rightarrow$ no precipitate. If $= K_{sp} \rightarrow$ saturated, no ppt.

20. — Key points for full credit:

- Unsaturated: ion product $< K_{sp} \rightarrow$ more solid can dissolve $\rightarrow$ no precipitate.
- Saturated: ion product $= K_{sp} \rightarrow$ equilibrium; dissolving rate = precipitation rate.
- Supersaturated: ion product $> K_{sp} \rightarrow$ excess ions precipitate until ion product $= K_{sp}$.
- Physically: in supersaturated solution, ions collide and form solid lattice; in unsaturated, solid continues dissolving.

21. — Key points for full credit:

- Common ion: an ion that appears in two dissolved salts in the same solution.
- Example: mixing NaCl and AgCl — Cl^- is the common ion.
- Presence of common ion shifts dissolution equilibrium LEFT (Le Châtelier) $\rightarrow$ less of the second salt dissolves.
- The less soluble salt precipitates until K_{sp} is re-established.
- Example from PPT: $\text{Na}_2\text{CO}_3 + \text{MgCO}_3$ mixture — CO_3^{2-} is the common ion.

ANSWER KEY — Quiz 5: Comprehensive Review

Multiple Choice

1. c. definition: forward rate = reverse rate AND concentrations stable (not necessarily equal)
2. b. $\text{H}_2\text{O}(\text{g}) + \text{C}(\text{s}) \rightleftharpoons \text{H}_2(\text{g}) + \text{CO}(\text{g})$: C is solid $\rightarrow$ excluded; H_2O is gas $\rightarrow$ included $\rightarrow$
 $K_{\text{eq}} = [\text{H}_2][\text{CO}] / [\text{H}_2\text{O}]$
3. d. temperature is the ONLY stress that changes the value of K_{eq}
4. c. precipitate forms when ion concentration product $> K_{\text{sp}}$
5. b. exothermic NH_3 : maximize yield with LOW temp + HIGH pressure + HIGH $[\text{N}_2]$ or $[\text{H}_2]$

True / False

6. True — concentrations are stable but not necessarily equal at equilibrium — fundamental definition
7. False — catalyst does NOT shift equilibrium; only speeds up approach to equilibrium
8. True — equal moles gas on both sides $\rightarrow$ pressure change has no net effect on shift
9. True — $K_{\text{eq}} > 1 \rightarrow$ forward predominates; $K_{\text{eq}} < 1 \rightarrow$ reverse predominates — PPT definition
10. False — pressure does NOT change K_{eq} — only temperature changes K_{eq}
11. True — solid is excluded from K_{sp} (concentration = 1, constant)
12. False — removing product $\rightarrow$ shift RIGHT (toward products) to replace what was removed
13. True — endothermic: heat is reactant; more heat $\rightarrow$ shift right $\rightarrow K_{\text{eq}}$ increases
14. True — ion product $< K_{\text{sp}} \rightarrow$ solution unsaturated $\rightarrow$ no precipitate forms
15. True — common ion shifts dissolution equilibrium left $\rightarrow$ less dissolves $\rightarrow$ decreased solubility

Short Answer

16.

Reaction: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ $\Delta H = -198 \text{ kJ}$ (exothermic)

a. K_{eq} expression: $K_{\text{eq}} = [\text{SO}_3]^2 / ([\text{SO}_2]^2[\text{O}_2])$

b. Pressure increased: left side = 3 mol gas, right side = 2 mol gas

Shift RIGHT (toward fewer moles of gas) ✓

c. Temperature increased (exothermic $\rightarrow$ heat is a product): shift LEFT, K_{eq} DECREASES

d. SO_3 removed: shift RIGHT to replace lost product ✓

17.

Reaction: $\text{CaF}_2(\text{s}) \rightleftharpoons \text{Ca}^{2+}(\text{aq}) + 2\text{F}^{-}(\text{aq})$

$K_{\text{sp}} = 3.9 \times 10^{-11}$

$[\text{Ca}^{2+}] = S$, $[\text{F}^{-}] = 2S$

$K_{\text{sp}} = [\text{Ca}^{2+}][\text{F}^{-}]^2 = S(2S)^2 = 4S^3$

$4S^3 = 3.9 \times 10^{-11}$

$$S^3 = 3.9 \times 10^{-11} / 4 = 9.75 \times 10^{-12}$$

$$S = (9.75 \times 10^{-12})^{1/3}$$

$$\text{Solubility} \approx 2.14 \times 10^{-4} \text{ mol/L}$$

18.

Le Châtelier's principle: when a stress is placed on a reversible process at equilibrium, the system shifts in the direction that relieves that stress.

Industrial example: Haber process ($\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$).

Manufacturers use high pressure (shifts right, more product) and remove NH_3 continuously (shifts right by Le Châtelier) to maximize ammonia yield.

Moderate temperature with a catalyst balances rate vs. equilibrium yield.

Essay — Key Points

19. — Key points for full credit:

- 17.1 — Equilibrium: forward rate = reverse rate; concentrations stable. $K_{eq} = [\text{products}]/[\text{reactants}]$ with exponents from coefficients; solids/liquids excluded.
- 17.2 — Le Châtelier: 3 stresses: concentration (no K_{eq} change), pressure (no K_{eq} change), temperature (CHANGES K_{eq}).
- 17.3 — K_{sp} : equilibrium constant for dissolving. Compare ion product to K_{sp} to determine saturation. Common-ion effect decreases solubility.

20. — Key points for full credit:

- $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$ is exothermic (3 mol gas $\rightarrow$ 2 mol gas).
- Temperature: use LOW temperature $\rightarrow$ favors forward (exothermic), higher K_{eq} . Trade-off: slower rate.
- Pressure: use HIGH pressure $\rightarrow$ shifts toward fewer moles (products). No K_{eq} change.
- Concentration: add excess SO_2 or O_2 to push equilibrium right.
- Remove SO_3 as it forms to continuously drive equilibrium right.
- Use a catalyst (e.g., V_2O_5) to reach equilibrium faster without affecting K_{eq} .

21. — Key points for full credit:

- Similarities: both are equilibrium constants; both are written products/reactants; both exclude solids.
- Differences: K_{eq} applies to any equilibrium; K_{sp} specifically applies to dissolving ionic solids.
- K_{eq} used to: predict direction of reaction, calculate equilibrium concentrations.
- K_{sp} used to: determine solubility, predict precipitation, assess saturation.
- K_{eq} example: $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$, $K_{eq} = [\text{NH}_3]^2/([\text{N}_2][\text{H}_2]^3) \approx 3.6 \times 10^8$ at given conditions.
- K_{sp} example: $\text{AgCl} \rightleftharpoons \text{Ag}^+ + \text{Cl}^-$, $K_{sp} = 1.8 \times 10^{-10} \rightarrow S = \sqrt{K_{sp}} = 1.34 \times 10^{-5} \text{ mol/L}$.